



Clear Churn

ANALYTICS

Turning Customer Exits into Business Intelligence

CUSTOMER CHURN ANALYSIS

A Cross-Industry Investigation: Telecom × Banking

"Turning Customer Exits into Business Intelligence"

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The Business Problem

Every year, subscription businesses lose millions to a problem they can predict


1 in 4

Telecom customers
cancels every year


\$1.67M

Lost annually
by one company


60.4%

Highest-risk segment
churn rate

ClearChurn Analytics was engaged to identify who is leaving, why they are leaving, and what can be done about it, across two industries simultaneously.

FACT: It costs 5-7× more to acquire a new customer than to keep an existing one



Acquiring new customers is like spending money on a flashy dating app. Retaining them is the actual marriage. It requires wayless budget and a lot more active listening.

Our Approach

A two-industry, data-driven investigation

DATASET 1



Telecom

7,032 customers

21 variables

IBM Telco Dataset

Primary analysis

METHODOLOGY



Python + Pandas

SQL — MySQL

EDA + Statistics

Tableau Dashboards

Z · T · Chi² tests

DATASET 2



Banking

10,000 customers

13 variables

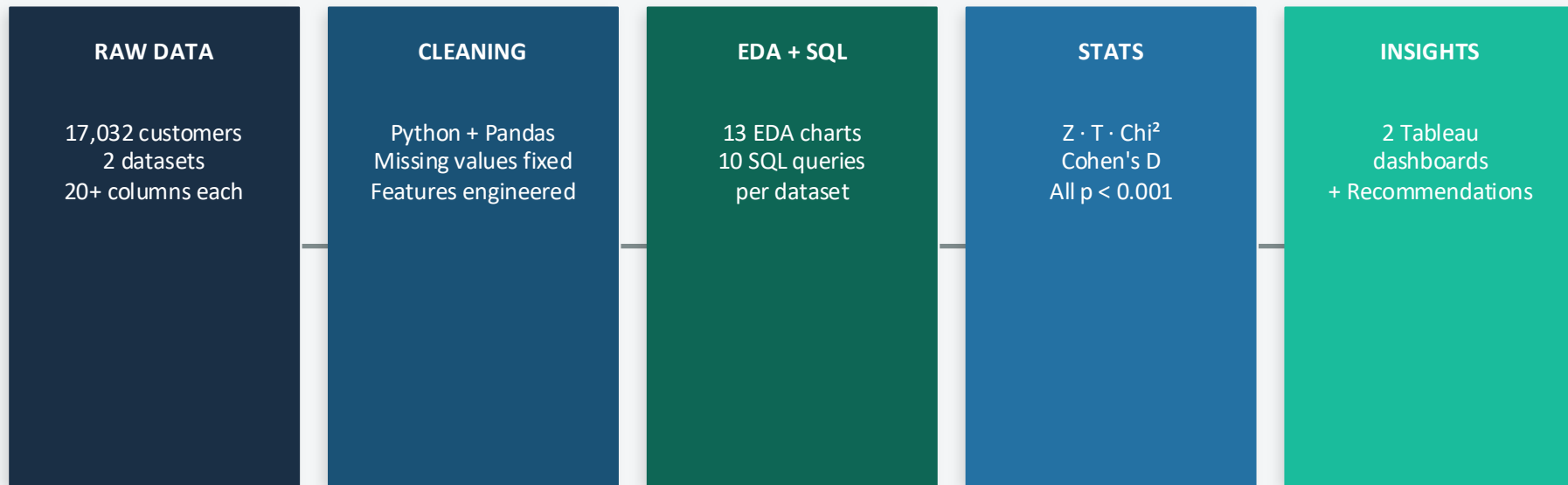
Kaggle Bank Dataset

Cross-industry

By analysing two industries simultaneously, we identify universal patterns versus industry-specific drivers

Data Pipeline

From raw data to actionable insights



Tools used: Python · Pandas · MySQL Workbench · Tableau Public · SciPy



We cleaned 17,032 rows of data, wrote 20 SQL queries, ran 7 statistical tests, and built 19 Tableau charts. All to answer one question: why do customers leave?

Telecom Churn Analysis

Live Tableau Dashboard — Interactive Deep Dive

[TABLEAU DASHBOARD 1]
Switch to Tableau at this point

01

Contract type is everything — 15× difference M2M vs two-year

02

The danger zone — first 12 months critical, 47.7% churn

03

Revenue impact — \$1.67M annually, 87% from M2M contracts

Cross-Industry Churn Comparison

Live Tableau Dashboard — Telecom vs Banking

[TABLEAU DASHBOARD 2]
Switch to Tableau at this point

01

Both industries lose high-value customers first — proven statistically

02

Engagement protects in both — support/activity halves churn rate

03

Key difference — tenure predicts Telecom churn, irrelevant in Banking

Statistical Validation



Every key finding confirmed — these are not coincidences

Test	Finding	Statistic	Result
Z-test	Telco vs Bank churn rate difference	$z = 9.49$	✓ $p < 0.001$
T-test	Churners pay/hold more in both industries	$t = 16.48$	✓ $p < 0.001$
Chi-square	Contract type drives Telecom churn	$\chi^2 = 1179$	✓ $p < 0.001$
Chi-square	Geography drives Banking churn	$\chi^2 = 301$	✓ $p < 0.001$
Correlation	Tenure predicts churn — Telecom	$r = -0.35$	✓ $p < 0.001$
Correlation	Tenure irrelevant — Banking	$r = -0.01$	✗ $p = 0.16$
Cohen's D	Tenure effect size — Telecom	$d = 0.86$	Large effect



$p < 0.001$ means there is less than 0.1% chance any of this is random luck. In plain English: we are 99.9% confident these findings are real.

Business Recommendations

What should these companies actually do?

TELECOM RECOMMENDATIONS



Convert Month to Month contract type customers with loyalty incentives at months 3, 6, and 12



Migrate electronic check users to automatic payment



Bundle tech support into onboarding . It cuts churn from 49% to 9%



Flag high-risk triple: Month to Month + Fibre + E-check = 60.4% churn

BANKING RECOMMENDATIONS



Target 41–60 age group with dedicated retention campaigns



Investigate Germany — 32.4% vs 16% elsewhere



Push customers from 1 to 2 products — churn drops from 27% to 7.6%



Re-engage inactive members before they leave

Conclusions & Future Work

What we learned — and what comes next

KEY CONCLUSIONS

- ✓ Churn is predictable — not inevitable
- ✓ Both industries lose high-value customers first
- ✓ Engagement protects customers in both industries
- ✓ Different industries need different strategies
- ✓ Tenure matters in Telecom — irrelevant in Banking

WHAT COULD BE IMPROVED

- Real-time data integration pipeline
- Machine learning churn prediction model
- Customer lifetime value analysis
- Qualitative research , exit interviews
- Expand to third industry , retail or SaaS

"Churn is not inevitable — it is predictable. The data does not just tell us who left. It tells us who is about to leave."



Clear Churn

ANALYTICS

Turning Customer Exits into Business Intelligence

Thank You



Churn rate of this presentation: 0% — you made it to the end!

Presented by: Hina Haq — Data Analytics Consultant

*"Turning Customer Exits
into Business Intelligence"*